

BobaLink Browser Extension “ Developer Guide

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Audience: Contributors, Fork Maintainers, Advanced Users
Companion Documents: [Technical Specification](#) | [API Reference](#) | [Installation Guide](#)

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1. Development Environment Setup

1.1 Prerequisites

Before you begin, ensure your development machine meets the following requirements:

Requirement	Minimum Version	Recommended	Purpose
Node.js	18 LTS	20 LTS	Runtime and package management
npm	9.x	10.x	Package manager (bundled with Node.js)
Git	2.40	2.43+	Version control
A supported browser	Chrome 88+	Chrome 120+	Testing and debugging

Verify your environment:

```
node --version    # Should print v18.x.x or higher
npm --version     # Should print 9.x.x or higher
git --version     # Should print 2.40+ or higher
```

1.2 Cloning the Repository

```
# Clone the repository
git clone https://github.com/bobaproject/bobalink.git

# Navigate to the project directory
```

```
cd bobalink
```

```
# Install dependencies
npm install
```

The `npm install` command installs all development and runtime dependencies. This may take 2â€“5 minutes depending on your connection speed.

1.3 Installing Dependencies

Key dependencies that will be installed:

Package	Version	Purpose
wxt	^0.17	Extension build framework
typescript	^5.0	Type safety
vite	^5.0	Module bundler (via WXT)
jest	^29.0	Unit testing framework
@playwright/test	^1.40	E2E testing framework
eslint	^8.0	Code linting
prettier	^3.0	Code formatting
bencode-js	^3.0	Bencode parsing for .torrent files
lz-string	^1.5	Compression for queue storage
webextension-polyfill	^0.10	Firefox compatibility

1.4 IDE Recommendations

Visual Studio Code (Recommended)

VS Code provides the best development experience for BobaLink. Recommended configuration:

`.vscode/settings.json`:

```
{
  "editor.formatOnSave": true,
  "editor.defaultFormatter": "esbenp.prettier-vscode",
  "editor.codeActionsOnSave": {
    "source.fixAll.eslint": "explicit"
  },
  "typescript.tsdk": "node_modules/typescript/lib",
  "typescript.preferences.importModuleSpecifier": "relative",
  "files.exclude": {
    "**/node_modules": true,
    "**/.output": true,
    "**/dist": true
  },
  "search.exclude": {
    "**/node_modules": true,
    "**/dist": true,
    "**/.output": true,
    "**/*.lock": true
  }
}
```

```

    }
  }
}

.vscode/extensions.json:

{
  "recommendations": [
    "esbenp.prettier-vscode",
    "dbaeumer.vscode-eslint",
    "ms-vscode.vscode-typescript-next",
    "bradlc.vscode-tailwindcss",
    "ryanluker.vscode-coverage-gutters",
    "orta.vscode-jest"
  ]
}

```

Recommended Extensions:

Extension	Publisher	Purpose
Prettier	esbenp.prettier-vscode	Code formatting
ESLint	dbaeumer.vscode-eslint	Real-time linting
Tailwind CSS IntelliSense	bradlc.vscode-tailwindcss	CSS autocomplete
Coverage Gutters	ryanluker.vscode-coverage-gutters	Visual test coverage
Jest	orta.vscode-jest	Test runner integration

WebStorm / IntelliJ IDEA

If using JetBrains IDEs: 1. Enable the Prettier plugin: **Settings** → **Languages & Frameworks** → **JavaScript** → **Prettier**. 2. Enable ESLint: **Settings** → **Languages & Frameworks** → **JavaScript** → **Code Quality Tools** → **ESLint**. 3. Set TypeScript version: **Settings** → **Languages & Frameworks** → **TypeScript** → point to `node_modules/typescript/lib`.

2. Project Structure

2.1 Directory Layout

```

bobalink/
├── src/
│   ├── assets/                # Static assets (icons, images)
│   │   ├── icon-16.png
│   │   ├── icon-32.png
│   │   ├── icon-48.png
│   │   └── icon-128.png
│   ├── background/           # Service worker (MV3 background scri
│   │   └── index.ts          # Entry point
│   ├── message-router.ts
│   ├── init.ts
│   └── content-scripts/      # Scripts injected into web pages

```

```

â",    â",    â"€â"€â"€ index.ts          # Entry point
â",    â",    â"€â"€â"€ magnet-scanner.ts
â",    â",    â"€â"€â"€ dom-observer.ts
â",    â",    â""â"€â"€ torrent-parser.ts
â",    â"€â"€â"€ popup/                # Toolbar popup UI
â",    â",    â"€â"€â"€ index.html
â",    â",    â"€â"€â"€ index.ts
â",    â",    â"€â"€â"€ popup.css
â",    â",    â""â"€â"€ components/
â",    â",    â"€â"€â"€ torrent-list.ts
â",    â",    â"€â"€â"€ queue-view.ts
â",    â",    â"€â"€â"€ status-bar.ts
â",    â",    â""â"€â"€ tab-groups.ts
â",    â"€â"€â"€ options/                # Options/settings page
â",    â",    â"€â"€â"€ index.html
â",    â",    â"€â"€â"€ index.ts
â",    â",    â"€â"€â"€ options.css
â",    â",    â""â"€â"€ components/
â",    â",    â"€â"€â"€ server-config.ts
â",    â",    â"€â"€â"€ download-prefs.ts
â",    â",    â"€â"€â"€ queue-settings.ts
â",    â",    â""â"€â"€ ui-settings.ts
â",    â"€â"€â"€ modules/                # Shared business logic modules
â",    â",    â"€â"€â"€ api-client.ts
â",    â",    â"€â"€â"€ auth-manager.ts
â",    â",    â"€â"€â"€ badge-manager.ts
â",    â",    â"€â"€â"€ config-manager.ts
â",    â",    â"€â"€â"€ crypto-utils.ts
â",    â",    â"€â"€â"€ discovery-service.ts
â",    â",    â"€â"€â"€ health-monitor.ts
â",    â",    â"€â"€â"€ i18n.ts
â",    â",    â"€â"€â"€ notification-manager.ts
â",    â",    â"€â"€â"€ queue-manager.ts
â",    â",    â"€â"€â"€ rate-limiter.ts
â",    â",    â"€â"€â"€ state-manager.ts
â",    â",    â""â"€â"€ tab-manager.ts
â",    â"€â"€â"€ types/                    # TypeScript type definitions
â",    â",    â"€â"€â"€ index.ts
â",    â",    â"€â"€â"€ messages.ts
â",    â",    â"€â"€â"€ torrent.ts
â",    â",    â"€â"€â"€ config.ts
â",    â",    â""â"€â"€ api.ts
â",    â""â"€â"€ utils/                # Utility functions
â",    â"€â"€â"€ bencode.ts
â",    â"€â"€â"€ hash.ts
â",    â"€â"€â"€ magnet-uri.ts
â",    â"€â"€â"€ storage.ts
â",    â"€â"€â"€ time.ts
â",    â""â"€â"€ validation.ts
â"€â"€â"€ tests/
â",    â"€â"€â"€ unit/                # Jest unit tests
â",    â",    â"€â"€â"€ modules/
â",    â",    â"€â"€â"€ utils/
â",    â",    â""â"€â"€ setup.ts

```

```

â",    â"œâ"€â"€ e2e/                    # Playwright E2E tests
â",    â",    â"œâ"€â"€ fixtures/
â",    â",    â"œâ"€â"€ specs/
â",    â",    â""â"€â"€ playwright.config.ts
â",    â""â"€â"€ mocks/                    # Test mocks
â",    â"œâ"€â"€ chrome-api.ts
â",    â""â"€â"€ fetch.ts
â"œâ"€â"€ public/                        # Public static files
â"œâ"€â"€ _locales/                      # Internationalization
â",    â"œâ"€â"€ en/
â",    â"œâ"€â"€ zh_CN/
â",    â"œâ"€â"€ es/
â",    â""â"€â"€ ...
â"œâ"€â"€ wxt.config.ts                  # WXT build configuration
â"œâ"€â"€ tsconfig.json                  # TypeScript configuration
â"œâ"€â"€ jest.config.ts                  # Jest test configuration
â"œâ"€â"€ eslint.config.js                # ESLint configuration
â"œâ"€â"€ prettier.config.js              # Prettier configuration
â"œâ"€â"€ package.json                    # Project manifest
â""â"€â"€ README.md

```

2.2 Module Descriptions

Background (Service Worker)

Module	Responsibility
index.ts	Service worker entry point; registers event listeners
message-router.ts	Routes incoming extension messages to appropriate handlers
init.ts	Extension initialization: config loading, health monitoring startup

Content Scripts

Module	Responsibility
index.ts	Content script entry; initializes scanner and observer
magnet-scanner.ts	Scans DOM for magnet links and .torrent references
dom-observer.ts	MutationObserver wrapper for dynamic content detection
torrent-parser.ts	Parses magnet URIs and .torrent file metadata

Modules (Shared Logic)

Module	Responsibility
api-client.ts	HTTP client for Boba and qBitTorrent API calls
auth-manager.ts	

Module	Responsibility
	Authentication: login, token refresh, session management
badge-manager.ts	Toolbar badge state: color, text, animations
config-manager.ts	Configuration CRUD with validation and persistence
crypto-utils.ts	AES-256-GCM encryption, PBKDF2 key derivation
discovery-service.ts	mDNS-like local network server discovery
health-monitor.ts	Periodic health checks; connection state management
i18n.ts	Internationalization: message lookup, locale detection
notification-manager.ts	Browser notification creation and management
queue-manager.ts	Offline queue: enqueue, dequeue, retry scheduling
rate-limiter.ts	Token bucket rate limiter for API calls
state-manager.ts	Central state store; reactive updates to UI
tab-manager.ts	Tab group enumeration; batch scan orchestration

Types

Module	Responsibility
index.ts	Barrel export for all types
messages.ts	Extension message passing types
torrent.ts	TorrentInfo and related types
config.ts	ExtensionConfig and sub-configuration types
api.ts	API request/response types for Boba and qBitTorrent

3. Building from Source

3.1 Development Build

Start the development server with hot module replacement:

```
npm run dev
```

This command: 1. Starts WXT in development mode. 2. Watches source files for changes and rebuilds automatically. 3. Outputs to `.output/` directory. 4. Provides a Chrome-compatible build by default.

Output structure:

```
.output/
  â€”â€”â€” chrome-mv3-dev/      # Chrome development build
  â€”â€”â€” manifest.json
  â€”â€”â€” background.js
```

```
â”,    â”œâ”€â”€ content-scripts/
â”,    â”œâ”€â”€ popup/
â”,    â””â”€â”€ options/
```

Loading the development build:

Browser

Chrome

Firefox

Opera

Steps

```
chrome://extensions â†’ Enable
âœœDeveloper modeâœœ â†’ âœœLoad
unpackedâœœ â†’ Select .output/chrome-
mv3-dev/
about:debugging â†’ âœœThis Firefoxâœœ
â†’ âœœLoad Temporary Add-onâœœ â†’ Select
manifest.json from .output/
firefox-mv3-dev/
opera://extensions â†’ Enable
âœœDeveloper modeâœœ â†’ âœœLoad
unpackedâœœ
```

3.2 Production Build

Create optimized production builds for all targets:

```
npm run build
```

This generates:

```
.output/
â”œâ”€â”€ chrome-mv3-prod/    # Chrome production build
â”œâ”€â”€ firefox-mv3-prod/   # Firefox production build
â”œâ”€â”€ opera-mv3-prod/     # Opera production build
â””â”€â”€ edge-mv3-prod/      # Edge production build (same as Chrome)
```

Each build is: - Minified and tree-shaken. - Source-mapped (.map files). - Ready for packaging as ZIP for store submission.

3.3 Cross-Browser Builds

Build for specific browsers:

```
# Chrome only
npx wxt build -b chrome
```

```
# Firefox only
npx wxt build -b firefox
```

```
# Opera only
npx wxt build --browser opera
```

Browser-specific differences are handled by: - WXTâ€™s browser-specific entry points (popup/index.firefox.ts). - Conditional compilation via import.meta.browser. - The webextension-polyfill package for API normalization.

3.4 Build Configuration

wxt.config.ts:

```
import { defineConfig } from 'wxt';

export default defineConfig({
  srcDir: 'src',
  outDir: '.output',
  modules: ['@wxt-dev/module-typescript'],
  manifest: {
    name: '__MSG_extName__',
    description: '__MSG_extDescription__',
    version: '1.0.0',
    permissions: ['activeTab', 'storage', 'contextMenus', 'notifications'],
    host_permissions: ['https://*/'],
    action: {
      default_popup: 'popup/index.html'
    },
    options_ui: {
      page: 'options/index.html',
      open_in_tab: true
    },
    commands: {
      'send-current-page': {
        suggested_key: { default: 'Ctrl+Shift+B', mac: 'Command+Shift+B' },
        description: 'Send all torrents on current page'
      }
    }
  }
});
```

4. Testing

4.1 Running Unit Tests

Execute the full unit test suite:

```
npm test
```

Or with Jest directly for more control:

```
# Run all tests
npx jest
```

```
# Run with coverage
npx jest --coverage
```

```
# Run in watch mode (during development)
npx jest --watch
```

```
# Run a specific test file
```



```
npx jest tests/unit/utis/magnet-uri.test.ts
```

```
# Run tests matching a pattern
```

```
npx jest --testNamePattern="parseMagnetUri"
```

Coverage reporting:

Coverage reports are generated in `coverage/`:

coverage/

```
â”€â”€ lcov-report/      # HTML report (open index.html in browser)
```

```
â"€â"€â"€ lcov.info # LCOV format for CI integration
```

coverage-summary.json

Coverage thresholds (enforced in CI):

Metric	Minimum
--------	---------

Statements 80%

Branches 75%

Functions 80%

Lines 80%

4.2 Running E2E Tests

End-to-end tests use Playwright to automate real browser interactions:

Install Playwright browsers (one-time setup)

```
npx playwright install chromium firefox
```

```
# Run all E2E tests
```

```
npm run test:e2e
```

```
# Run in headed mode (visible browser)
```

```
npx playwright test --headed
```

```
# Run a specific spec
```

```
npx playwright test tests/e2e/specs/detection.spec.ts
```

```
# Run with UI mode (interactive debugging)
```

```
npx playwright test --ui
```

```
# Run on specific browser
```

```
npx playwright test --project=chromium
```

n

E2E Test Structure:

```
tests/e2e/ â”€â”€â”€ fixtures/ # Test data â”€, â”€â”€â”€â”€ sample.torrent â”€, â”€â”€â”€â”€ magnet-
links.html â”€, â”€â”€â”€â”€ mock-server.ts â”€â”€â”€â”€ specs/ # Test specifications â”€, â”€â”€â”€â”€
detection.spec.ts # Torrent detection tests â”€, â”€â”€â”€â”€ sending.spec.ts # Send-to-server tests
â”€, â”€â”€â”€â”€ queue.spec.ts # Queue management tests â”€, â”€â”€â”€â”€ auth.spec.ts #
Authentication tests â”€, â”€â”€â”€â”€ settings.spec.ts # Options page tests â”€â”€â”€â”€ page-objects/
```

```
# Page Object Models â”, â”œâ”€â”€ popup.page.ts â”, â”œâ”€â”€ options.page.ts â”, â””â”€â”€ mock-server.page.ts â””â”€â”€ playwright.config.ts # Playwright configuration
```

4.3 Writing New Tests

Unit Test Template

```
````typescript
import { parseMagnetUri } from '@src/utils/magnet-uri';

describe('parseMagnetUri', () => {
 it('should parse a complete magnet URI', () => {
 const uri = 'magnet:?xt=urn:btih:abc123def456&dn=Test+File&tr=https://';
 const result = parseMagnetUri(uri);

 expect(result.infohash).toBe('abc123def456');
 expect(result.displayName).toBe('Test File');
 expect(result.trackers).toContain('https://tracker.example.com/announce');
 });

 it('should throw on invalid magnet URI', () => {
 expect(() => parseMagnetUri('not-a-magnet')).toThrow('Invalid magnet URI');
 });

 it('should handle missing display name', () => {
 const uri = 'magnet:?xt=urn:btih:abc123def456';
 const result = parseMagnetUri(uri);
 expect(result.displayName).toBe('Unknown Torrent');
 });
});
```

#### E2E Test Template

```
import { test, expect } from '@playwright/test';
import { PopupPage } from '../page-objects/popup.page';

test.describe('Torrent Detection', () => {
 test('should detect magnet links on a page', async ({ page, extensionId }) {
 // Arrange: Load a test page with magnet links
 await page.goto(`file://${__dirname}/../fixtures/magnet-links.html`);

 // Act: Open the extension popup
 const popup = new PopupPage(page, extensionId);
 await popup.open();

 // Assert: Verify torrents are listed
 const torrentCount = await popup.getTorrentCount();
 expect(torrentCount).toBe(3);

 const firstTorrentName = await popup.getTorrentName(0);
 expect(firstTorrentName).toContain('Ubuntu');
 });
});
```

```
 });
 });
```

## 4.4 Test Coverage Reports

Generate and view coverage:

```
npm run test:coverage
```

The HTML report opens automatically. Key files to monitor:

File	Target	Notes
src/modules/api-client.ts	90%	Critical for reliability
src/modules/crypto-utils.ts	95%	Security-critical
src/modules/queue-manager.ts	85%	Core functionality
src/content-scripts/magnet-scanner.ts	80%	User-facing
src/utils/magnet-uri.ts	95%	Pure functions, easy to test

---

## 5. Code Style and Conventions

### 5.1 TypeScript Strict Mode

The project uses TypeScript strict mode. Key rules:

```
{
 "compilerOptions": {
 "strict": true,
 "noImplicitAny": true,
 "strictNullChecks": true,
 "strictFunctionTypes": true,
 "strictBindCallApply": true,
 "strictPropertyInitialization": true,
 "noImplicitReturns": true,
 "noFallthroughCasesInSwitch": true,
 "noUncheckedIndexedAccess": true
 }
}
```

**No any types in production code.** Use unknown with type guards:

```
// Bad
function process(data: any): any {
 return data.value;
}

// Good
function process(data: unknown): string {
 if (typeof data === 'object' && data !== null && 'value' in data) {
 return String((data as Record<string, unknown>).value);
 }
}
```

```
 throw new TypeError('Invalid data shape');
}
```

## 5.2 ESLint Configuration

```
// eslint.config.js
export default [
 {
 files: ['src/**/*.ts'],
 languageOptions: {
 parser: '@typescript-eslint/parser',
 parserOptions: {
 project: './tsconfig.json'
 }
 },
 plugins: {
 '@typescript-eslint': typescriptEslint
 },
 rules: {
 '@typescript-eslint/no-explicit-any': 'error',
 '@typescript-eslint/no-unused-vars': ['error', { argsIgnorePattern:
 '@typescript-eslint/explicit-function-return-type': 'warn',
 '@typescript-eslint/no-non-null-assertion': 'error',
 '@typescript-eslint/prefer-nullish-coalescing': 'error',
 '@typescript-eslint/prefer-optional-chain': 'error',
 'no-console': ['warn', { allow: ['error', 'warn'] }]
 }
 }
];
```

## 5.3 Prettier Configuration

```
// prettier.config.js
export default {
 semi: true,
 trailingComma: 'es5',
 singleQuote: true,
 printWidth: 100,
 tabWidth: 2,
 useTabs: false,
 bracketSpacing: true,
 arrowParens: 'avoid'
};
```

## 5.4 Naming Conventions

Construct	Convention	Example
Variables / Functions	camelCase	sendTorrent, getStatus
Constants	UPPER_SNAKE_CASE	MAX_QUEUE_SIZE, DEFAULT_TIMEOUT
Types / Interfaces	PascalCase	TorrentInfo, ServerConfig

Construct	Convention	Example
Classes	PascalCase	QueueManager, ApiClient
Enums	PascalCase + PascalCase members	enum ConnectionState { Connected, Disconnected }
Files	kebab-case	magnet-scanner.ts, crypto-utils.ts
Test files	*.test.ts or *.spec.ts	magnet-uri.test.ts
Private members	Leading underscore	_internalQueue, _timerId

## 5.5 Comment Style (JSDoc)

All public APIs must include JSDoc comments:

```
/**
 * Parses a magnet URI string into a structured TorrentInfo object.
 *
 * @param uri - The raw magnet URI (e.g., "magnet:?xt=urn:btih:abc123...")
 * @returns Parsed torrent metadata
 * @throws {ValidationError} If the URI format is invalid
 *
 * @example
 * ```typescript
 * const info = parseMagnetUri('magnet:?xt=urn:btih:abc123&dn=Example');
 * console.log(info.infohash); // 'abc123...'
 * ```
 */
export function parseMagnetUri(uri: string): TorrentInfo {
 // Implementation
}
```

Comment guidelines: - Explain **why**, not **what** (the code shows what). - Document all parameters, return values, and thrown errors. - Include usage examples for complex functions. - Use `TODO(username): description` for known technical debt. - Use `FIXME(username): description` for bugs that need fixing.

---

## 6. Architecture Decisions

### 6.1 Why MV3 Over MV2

**Decision:** Use Manifest V3 (MV3) instead of Manifest V2 (MV2).

**Rationale:**

Factor	MV2	MV3	Winner
Chrome Web Store acceptance	Being deprecated	Required for new extensions	MV3
		Ephemeral service worker	MV3

Factor	MV2	MV3	Winner
Service worker vs background page	Persistent background page		
Performance	Background page always running	Service worker wakes on events	MV3
Memory usage	Higher (persistent)	Lower (ephemeral)	MV3
Network request modification	<code>webRequest</code> blocking	<code>declarativeNetRequest</code>	MV2*
Cross-browser support	Legacy only	Modern standard	MV3

\* BobaLink does not need to block/modify network requests, so the `webRequest` limitation of MV3 is not a factor.

**Mitigations for MV3 limitations:** - Service worker persistence: Use `chrome.alarms` API to keep the worker alive during active operations. - Storage: Use `chrome.storage` (local and session) instead of in-memory state. - All Chrome, Firefox, Opera, and Yandex versions targeted by BobaLink support MV3.

## 6.2 Why WXT Over Raw Webpack/Vite

**Decision:** Use WXT as the build framework instead of configuring Webpack or Vite manually.

**Rationale:**

Factor	Raw Vite	WXT	Winner
Extension-specific scaffolding	None	Built-in	WXT
HMR for content scripts	Manual config	Automatic	WXT
Multi-browser builds	Manual config	Automatic	WXT
Manifest generation	Manual	Auto-generated from config	WXT
File-based routing	None	Entry points auto-discovered	WXT
Dev server with auto-reload	Manual config	Built-in	WXT
Bundle size	Comparable	Comparable (thin wrapper)	Tie
Community size	Large	Growing	Vite

**WXT specifically solves:** - Automatic discovery of `src/popup/`, `src/options/`, `src/background/`, `src/content-scripts/` entries. - Browser-specific build variants (Chrome MV3, Firefox MV3 with polyfills). - Dev mode that auto-reloads the extension on file change. - Automatic `il8n` message placeholder substitution in the manifest.

## 6.3 Why activeTab Over host\_permissions

**Decision:** Use `activeTab` as the primary permission model instead of `host_permissions`.

**Rationale:**

`activeTab` grants temporary access to the current tab only when the user interacts with the extension. This is more privacy-respecting and requires fewer permission prompts at install time.

Permission Model	Install Prompt	Privacy	Functionality
<code>host_permissions: &lt;all_urls&gt;</code>	Scary â€œread all website dataâ€	Poor	Works on all pages
<code>activeTab</code>	No prompt (on-demand)	Excellent	Works on current tab only

**Implementation:** - Primary detection uses `activeTab` + user gesture (clicking the extension icon). - Optional `host_permissions` for advanced users who want proactive detection without interaction. - The `optional_permissions` field in the manifest allows users to grant additional access.

## 6.4 Why AES-256-GCM for Encryption

**Decision:** Use AES-256-GCM via the Web Crypto API for credential encryption.

**Rationale:**

Algorithm	Mode	Authentication	Performance	Standard
AES-128-CBC	No	No (requires HMAC)	Fast	Legacy
AES-256-CBC	No	No (requires HMAC)	Fast	Legacy
AES-256-GCM	Yes	Built-in	Fast	Modern standard
ChaCha20-Poly1305	Yes	Built-in	Fast	Alternative

AES-256-GCM was chosen because: 1. It provides authenticated encryption (confidentiality + integrity) in a single operation. 2. It is widely supported by the Web Crypto API in all targeted browsers. 3. It is a NIST standard with extensive cryptanalysis and hardware acceleration support. 4. ChaCha20-Poly1305 is excellent but less universally accelerated on x86 hardware.

**Key derivation:** - Master key derived from user password via PBKDF2 with 100,000 iterations. - Per-entry random IV (96-bit) generated via `crypto.getRandomValues()`. - 128-bit authentication tag validated on every decryption.

## 7. Contributing

### 7.1 Git Workflow (GitHub Flow)

BobaLink follows the GitHub Flow branching model:

```
graph LR
 A[main branch] --> B[Create feature branch]
 B --> C[Make commits]
 C --> D[Open Pull Request]
 D --> E[Code Review]
 E --> F[Merge to main]
 F --> G[Deploy]
```

**Branch naming conventions:**

Prefix	Purpose	Example
<code>feature/</code>	New features	<code>feature/tab-group-batch</code>

Prefix	Purpose	Example
bugfix/	Bug fixes	bugfix/badge-color-update
hotfix/	Critical production fixes	hotfix/auth-race-condition
docs/	Documentation updates	docs/api-examples
refactor/	Code refactoring	refactor/queue-manager
test/	Test additions/improvements	test/queue-coverage

## 7.2 Commit Message Conventions (Conventional Commits)

All commits must follow the Conventional Commits specification:

<type>(<scope>): <description>

[optional body]

[optional footer(s)]

### Types:

Type	Use When	Example
feat	Adding a new feature	feat(popup): add tab group selection UI
fix	Fixing a bug	fix(queue): resolve race condition on retry
docs	Documentation changes	docs(api): add qBitTorrent sync endpoint
style	Code formatting (no logic change)	style(lint): fix trailing commas
refactor	Code restructuring	refactor(auth): extract token refresh logic
test	Adding or updating tests	test(queue): add retry scheduling tests
chore	Build/tooling changes	chore(deps): update wxt to 0.17.0
perf	Performance improvements	perf(scanner): debounce mutation observer
security	Security-related changes	security(crypto): increase PBKDF2 iterations

### Scopes:

Scope	Description
popup	Popup UI
options	Options page



Scope	Description
content	Content scripts
background	Service worker
api	API client
queue	Queue manager
auth	Authentication
crypto	Encryption utilities
i18n	Internationalization
build	Build system
deps	Dependencies

### Breaking changes:

Append ! after the type/scope and include BREAKING CHANGE: in the footer:

```
feat(api)!: remove v1 auth endpoint
```

BREAKING CHANGE: The /api/v1/auth/login endpoint has been removed.  
Use /api/v2/auth/token instead.

## 7.3 Pull Request Template

When opening a pull request, use this template:

### ## Description

Brief description of the change and its purpose.

### ## Type of Change

- [ ] Bug fix
- [ ] New feature
- [ ] Breaking change
- [ ] Documentation update
- [ ] Refactoring

### ## Testing

- [ ] Unit tests added/updated
- [ ] E2E tests added/updated
- [ ] Manual testing performed

### ## Checklist

- [ ] Code follows project style guidelines
- [ ] Self-review completed
- [ ] JSDoc comments added for public APIs
- [ ] No `any` types introduced
- [ ] TypeScript compiles without errors
- [ ] ESLint passes without warnings
- [ ] Test coverage meets thresholds

### ## Related Issues

Fixes #123

Relates to #456

## 7.4 Code Review Checklist

Reviewers should verify:

Check	Description
Functionality	Does the change do what it claims?
Tests	Are there adequate tests? Do they pass?
Edge cases	Are error paths and boundary conditions handled?
Security	Does the change introduce any security risks?
Performance	Could this cause performance issues at scale?
i18n	Are all user-facing strings externalized?
Accessibility	Are ARIA labels and keyboard navigation present?
Type safety	Are there no any types or unchecked nulls?
Documentation	Is JSDoc updated? Is the user guide updated if needed?

## 7.5 Release Process

```
graph TD
 A[Accumulate features on main] --> B[Create release branch]
 B --> C[Update version in manifest]
 C --> D[Update CHANGELOG.md]
 D --> E[Run full test suite]
 E --> F{Tests pass?}
 F -->|No| G[Fix issues] --> E
 F -->|Yes| H[Create Git tag]
 H --> I[Build all targets]
 I --> J[Create GitHub Release]
 J --> K[Submit to Chrome Web Store]
 J --> L[Submit to AMO]
 J --> M[Submit to Opera Addons]
```

**Version numbering** follows Semantic Versioning (SemVer):

Bump	When	Example
Major (X.0.0)	Breaking API changes	2.0.0
Minor (x.Y.0)	New features, backward compatible	1.1.0
Patch (x.y.Z)	Bug fixes, no API changes	1.0.1

---

## 8. Extending the Extension

### 8.1 Adding New Torrent Site Selectors

Some torrent sites use non-standard HTML structures. You can add custom selectors:

1. Create a new file in `src/content-scripts/selectors/`:

```
// src/content-scripts/selectors/nyaa-si.ts
import { SiteSelector } from '@src/types/selector';

export const nyaaSiSelector: SiteSelector = {
 // Match against hostname
 hostname: /nyaa\.si$/,

 // CSS selector for magnet links
 magnetSelector: 'a[href^="magnet:?"]',

 // CSS selector for .torrent file links
 torrentSelector: 'a[href$=".torrent"]',

 // Optional: Extract display name from a different element
 nameExtractor: (linkElement: HTMLAnchorElement): string | null => {
 const row = linkElement.closest('tr');
 if (row) {
 const titleCell = row.querySelector('td:nth-child(2) a');
 return titleCell?.textContent?.trim() || null;
 }
 return null;
 },

 // Optional: Extract size from page
 sizeExtractor: (linkElement: HTMLAnchorElement): number | null => {
 const row = linkElement.closest('tr');
 if (row) {
 const sizeCell = row.querySelector('td:nth-child(4)');
 return parseSizeString(sizeCell?.textContent || '');
 }
 return null;
 },

 // Optional: Custom scan logic
 customScanner?: (root: HTMLElement): Partial<TorrentInfo>[] => {
 // Advanced: scan page-specific structures
 }
};
```

1. Register the selector in `src/content-scripts/selectors/index.ts`:

```
import { nyaaSiSelector } from './nyaa-si';

export const siteSelectors: SiteSelector[] = [
 // ... existing selectors
 nyaaSiSelector,
];
```

1. Write a test:

```
// tests/unit/content-scripts/selectors/nyaa-si.test.ts
import { nyaaSiSelector } from '@src/content-scripts/selectors/nyaa-si';

describe('nyaaSiSelector', () => {
```

```

it('should match nyaa.si hostname', () => {
 expect(nyaaSiSelector.hostname.test('nyaa.si')).toBe(true);
 expect(nyaaSiSelector.hostname.test('sukebei.nyaa.si')).toBe(true);
});

it('should extract name from table row', () => {
 document.body.innerHTML = `
 <table>
 <tr>
 <td></td>
 <td><a>My Torrent Title</td>
 <td></td>
 <td>1.2 GiB</td>
 <td>Magnet</td>
 </tr>
 </table>
 `;
 const link = document.querySelector('a[href^="magnet"]') as HTMLAnchorElement;
 expect(nyaaSiSelector.nameExtractor?.(link)).toBe('My Torrent Title');
});
});

```

## 8.2 Adding New Authentication Methods

To add a new authentication method (e.g., OAuth2):

1. Add the method to the AuthMethod union type:

```

// src/types/config.ts
type AuthMethod = 'cookie' | 'api-key' | 'basic' | 'custom-header' | 'oauth2';

```

1. Implement the auth handler:

```

// src/modules/auth/oauth2-handler.ts
import { AuthHandler, AuthCredentials } from '@src/types/auth';

export class OAuth2Handler implements AuthHandler {
 private token: string | null = null;
 private refreshToken: string | null = null;
 private expiresAt = 0;

 async authenticate(credentials: AuthCredentials): Promise<void> {
 const response = await fetch(`${credentials.serverUrl}/oauth/token`, {
 method: 'POST',
 headers: { 'Content-Type': 'application/json' },
 body: JSON.stringify({
 grant_type: 'password',
 client_id: credentials.clientId,
 username: credentials.username,
 password: credentials.password,
 }),
 });

 if (!response.ok) {

```

```

 throw new AuthError('OAuth2 authentication failed', { cause: response.json() });
 }

 const data = await response.json();
 this.token = data.access_token;
 this.refreshToken = data.refresh_token;
 this.expiresAt = Date.now() + data.expires_in * 1000;
}

async getAuthHeaders(): Promise<Record<string, string>> {
 if (!this.token || Date.now() >= this.expiresAt) {
 await this.refresh();
 }
 return { Authorization: `Bearer ${this.token}` };
}

private async refresh(): Promise<void> {
 // Implementation
}
}

```

1. Register the handler:

```

// src/modules/auth-manager.ts
import { OAuth2Handler } from '../auth/oauth2-handler';

const authHandlers: Record<AuthMethod, () => AuthHandler> = {
 'cookie': () => new CookieHandler(),
 'api-key': () => new ApiKeyHandler(),
 'basic': () => new BasicAuthHandler(),
 'custom-header': () => new CustomHeaderHandler(),
 'oauth2': () => new OAuth2Handler(), // Add this line
};

```

1. Update the Options UI to include OAuth2 configuration fields.

## 8.3 Adding New Keyboard Shortcuts

To add a new keyboard shortcut:

1. Define the command in the manifest:

```

{
 "commands": {
 "send-current-page": {
 "suggested_key": { "default": "Ctrl+Shift+B" },
 "description": "Send all torrents on current page"
 },
 "your-new-command": {
 "suggested_key": { "default": "Ctrl+Shift+N" },
 "description": "Description of what it does"
 }
 }
}

```

```
}
}
```

1. Handle the command in the service worker:

```
// src/background/index.ts
chrome.commands.onCommand.addListener(async (command, tab) => {
 switch (command) {
 case 'send-current-page':
 await handleSendCurrentPage(tab);
 break;
 case 'your-new-command':
 await handleYourNewCommand(tab);
 break;
 }
});
```

1. Implement the handler:

```
// src/background/command-handlers.ts
export async function handleYourNewCommand(tab?: chrome.tabs.Tab): Promise<string> {
 if (!tab?.id) return;

 // Your implementation here
 const result = await chrome.tabs.sendMessage(tab.id, {
 type: 'YOUR_NEW_ACTION'
 });

 // Update badge or show notification
 await badgeManager.setStatus('processing');
}
```

1. Add i18n strings:

```
{
 "command_your_new_command": {
 "message": "Your new command description"
 }
}
```

---

## 9. Debugging

### 9.1 Chrome DevTools for Extensions

**Service Worker:** 1. Navigate to `chrome://extensions`. 2. Find BobaLink and click **Service Worker** (under **Inspect views**). 3. This opens DevTools dedicated to the service worker context. 4. Use the Console for logs, Network for API calls, and Sources for breakpoints.

**Content Script:** 1. Open the page where the content script is running. 2. Open DevTools (F12). 3. Go to the **Sources** panel. 4. In the file tree, find `chrome-extension://{extension-id}/`. 5. Set breakpoints in content script files.

**Popup:** 1. Click the BobaLink icon to open the popup. 2. Right-click inside the popup **Inspect**. 3. This opens a dedicated DevTools window for the popup.

#### Extension ID:

Mode	Extension ID Location
------	-----------------------

Development	chrome://extensions BobaLink ID field
Production	Fixed ID from Chrome Web Store

## 9.2 Firefox about:debugging

1. Navigate to `about:debugging`.
2. Click **This Firefox**.
3. Find BobaLink under Extensions.
4. Click **Inspect** to open DevTools.
5. Use Browser Console (`Ctrl+Shift+J`) for all extension logs.

## 9.3 Viewing Service Worker Logs

The service worker logs extensively for debugging. Log levels:

```
// src/utils/logger.ts
enum LogLevel {
 DEBUG = 0, // Verbose internal state
 INFO = 1, // Normal operations
 WARN = 2, // Recoverable issues
 ERROR = 3, // Failures requiring attention
}
```

#### Enable debug logging:

1. Open the service worker DevTools (see 9.1).
2. In the Console, run:  

```
chrome.storage.local.set({ 'debugMode': true });
```
3. Reload the service worker from `chrome://extensions`.
4. Debug logs will appear in the Console.

#### Log output examples:

```
[BobaLink] [INFO] Service worker initialized, version 1.0.0
[BobaLink] [DEBUG] Health check: GET https://boba.local:8443/health
[BobaLink] [INFO] Health check passed: Boba v1.4.2, latency 45ms
[BobaLink] [DEBUG] Magnet detected: abc123... on https://example.com
[BobaLink] [INFO] Torrent sent: abc123... (Ubuntu 24.04)
[BobaLink] [WARN] Send failed for def456... (Network timeout), queued for
[BobaLink] [ERROR] Auth failed: invalid API key
```

## 9.4 Inspecting Storage

Extension storage can be inspected via DevTools:

Storage Type	Chrome	Firefox
<code>chrome.storage.local</code>	DevTools â†’ Application â†’ Storage â†’ Extension Storage	DevTools â†’ Storage â†’ Extension Storage
<code>chrome.storage.session</code>	DevTools â†’ Application â†’ Storage â†’ Extension Storage (Session)	DevTools â†’ Storage â†’ Extension Storage

Viewing storage via console:

```
// View all stored data
chrome.storage.local.get(null, (data) => console.log(data));

// View configuration
chrome.storage.local.get('config', ({ config }) => console.log(config));

// View queue
chrome.storage.local.get('queue', ({ queue }) => console.log(queue));

// Clear all data (use with caution!)
chrome.storage.local.clear(() => console.log('Storage cleared'));
```

**Note:** Credential data is encrypted. You will see ciphertext, not plaintext passwords.

## 9.5 Common Debugging Scenarios

### Scenario: Content Script Not Injecting

**Symptoms:** No torrents detected on any page.

**Diagnostic steps:** 1. Check DevTools â†’ Console for injection errors. 2. Verify the content script is registered in `manifest.json`. 3. Check if the page has a restrictive CSP: `javascript // In page console document.querySelector('meta[http-equiv="Content-Security-Policy"]')?.content` 4. Check the service worker logs for injection errors.

### Scenario: API Calls Failing

**Symptoms:** â€œSend failedâ€ notifications, queue filling up.

**Diagnostic steps:** 1. Open service worker DevTools â†’ Network tab. 2. Trigger a send and watch the request. 3. Check request headers (especially `Referer` for qBitTorrent). 4. Check response status and body. 5. Verify CORS headers if cross-origin.

### Scenario: Queue Not Processing

**Symptoms:** Items stuck in queue, no retry attempts.



**Diagnostic steps:** 1. Check service worker is alive: `chrome://extensions/` Service Worker status. 2. Check for `chrome.alarms` registration: `javascript chrome.alarms.getAll((alarms) => console.log(alarms));` 3. Check queue processing logs in service worker console. 4. Manually trigger processing: `javascript chrome.runtime.sendMessage({ type: 'PROCESS_QUEUE' });`

## Scenario: Encryption Issues

**Symptoms:** “Cannot decrypt credentials” errors.

**Diagnostic steps:** 1. Check if the encryption key exists in session storage: `javascript chrome.storage.session.get('encryptionKey', (data) => { console.log('Key exists:', !!data.encryptionKey); });` 2. If the key is missing, the browser was likely restarted without password protection. 3. Re-enter credentials in Options to re-establish the encryption key.

---

*End of Developer Guide*